



Inspiring Excellence

CSE370 : Database Systems Project Report

Project Title : Urban Citizen Administration System

Group No : 04, CSE370 Lab Section : 11, Spring 2026		
ID	Name	Contribution
24101392	Fardin Islam	Backend, Frontend
24101331	Sadman Rahman	Backend, Frontend
24101328	Oyshi Sarker Nishita	Backend, Frontend

Table of Contents

Section No	Content	Page No
1	Introduction	3
2	Project Features	4
3	ER/EER Diagram	5
4	Schema Diagram	6
5	Normalization	7
6	Frontend Development	8-12
7	Backend Development	13
8	Source Code Repository	14
9	Conclusion	14
10	References	15

Introduction

[Briefly describe the idea of your project]

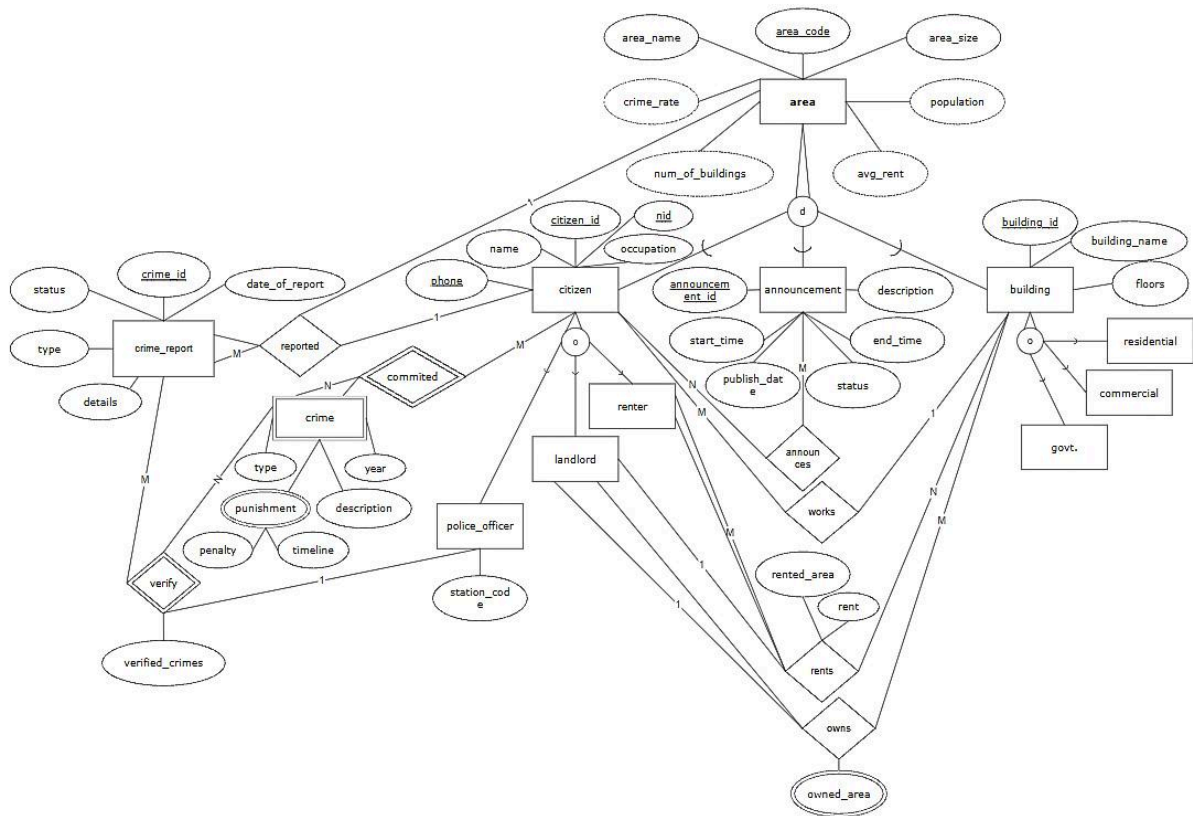
How is efficient city management possible without a centralized and dependable digital administrative system? While cities are becoming more modern, the process of handling datasets is still stuck in the past. The lack of coordinated communication between authorities, unstructured citizen records, scattered datasets, outdated crime reporting, etc., are still problems of the present. To handle these issues, we have developed the **Urban Citizen Administration System (UCAS)**, a web-based platform that centralizes key aspects of urban administration in a single secure system. It assigns various roles to different users to control their access, protecting privacy and data security.

The system provides features such as citizen and property management, crime reporting and verification, and a searchable criminal records database to assist law enforcement. It also has targeted announcements for better communication and a live analytics dashboard that offers insights into population data, rental trends, and crime rates. Built with PHP and MySQL, the platform has a structured database design that ensures data integrity and reliability. The City Management System aims to modernize city governance by improving transparency, public safety, and decision-making based on data.

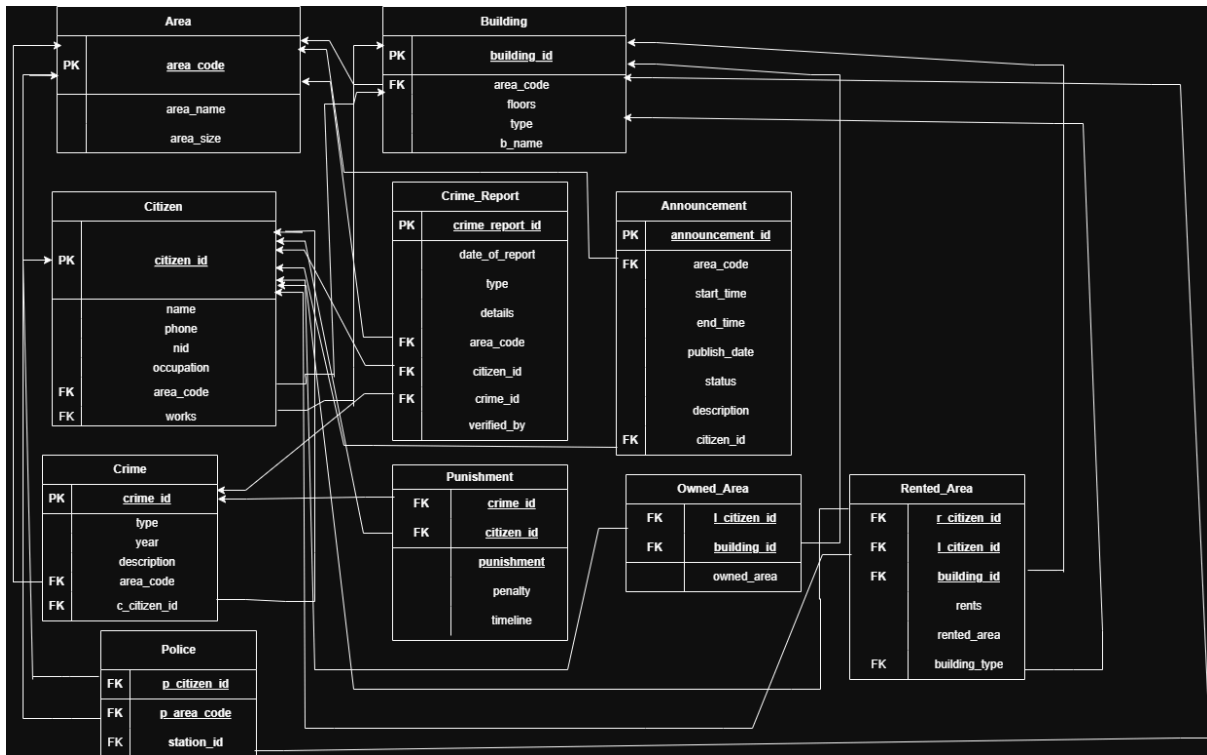
Project Features

ID, Name	Features [3 per member]	
24101392, Fardin Islam	Ft 1	Citizen Management [Register and manage citizen records with roles (Owner/Renter)]
	Ft 2	Area Management [Manage areas, buildings, and property ownership & renting]
	Ft 3	
24101331, Sadman Rahman	Ft 1	Tracking Criminal History [Create/update criminal history of citizens]
	Ft 2	Crime Reporting [Area-based crime reported by anonymous citizen & verified by police]
	Ft 3	
24101328, Oyshi Sarker Nishita	Ft 1	Announcement Management [Publish area and building announcements and notices]
	Ft 2	Area Data Analysis & Comparison [Generate report such as population, ownership, crime rate in each area and rent statistics]
	Ft 3	

ER/EER Diagram



Schema Diagram



Normalization

- a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
- b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
- c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?

ANSWER:

- a. **1NF:** The schema is in 1NF, since all the attributes are atomic and there are no repeating groups present.
- b. **2NF:** The database schema is in Second Normal Form since no non-prime attribute is dependent on a partial candidate key.
- c. **3NF:** The given database is in 3NF as there are no transitive dependencies. All non-prime attributes depend directly on the key attribute.

Frontend Development

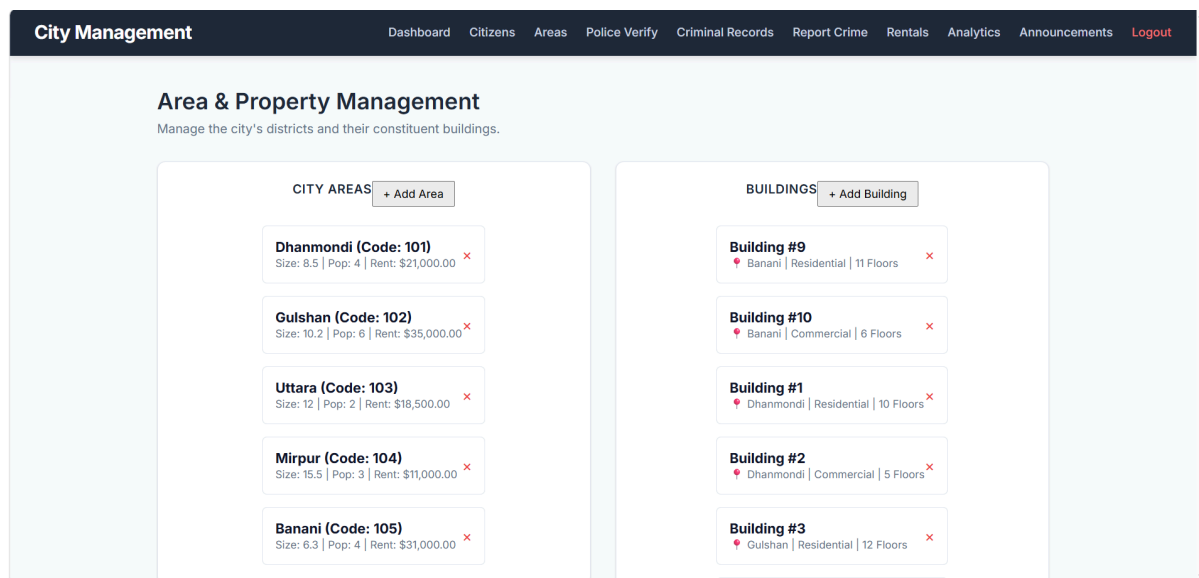
Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

The City Management System frontend application was created using HTML and CSS, keeping in mind a simple design for clarity and ease of use. This was done to ensure that users of the application (Admins, Owners, and Citizens) would be able to access various modules of the software without any hassle.

A variety of layout styles, like dashboards, tables, forms, and cards, were used for the proper representation of data. Progress bars for analytics and announcements in card format help better understand data. In other words, the frontend application serves as an interface between users and the software's backend.

Contribution of ID : 24101392, Name : Fardin Islam

- Citizen Management module (sign up and role selection)
- Area Management module (areas, buildings, ownership & renting interfaces)



City Management

DashboardCitizensAreasPolice VerifyCriminal RecordsReport CrimeRentalsAnalyticsAnnouncementsLogout

Citizen Management

Manage the city's registered citizens and roles.

+ Add Citizen

ID	Name	NID	Phone	Role	Area	Actions
21	Sadman Rahman	968	01631115975	Renter	Banani	EditDelete
20	Temp	NID230	01723232323	Renter	Gulshan	EditDelete
19	Taseen	NID120	01886404805	Police	Gulshan	EditDelete
17	Oyshi	NID100	0172996946116	Renter	Gulshan	EditDelete
16	System Admin	ADMIN	admin	Admin	Unassigned	EditDelete
15	Tania Sultana	NID015	01866666666	Admin	Banani	EditDelete
14	Shakib Khan	NID014	01855555555	Owner	Gulshan	EditDelete

City Management

DashboardCitizensAreasPolice VerifyCriminal RecordsReport CrimeRentalsAnalyticsAnnouncementsLogout

Rentals

Browse available homes for rent across the city.

+ Post Rental

House 5, Block C

Uttara (Bld. #5)

3 Bed, 3 Bath, Dining, Kitchen

\$25,000.00/moBy: Tania Sultana

localhost/city-management/pages/report_crime.php

© 2026 City Management System. All rights reserved.

City Management

DashboardCitizensAreasPolice VerifyCriminal RecordsReport CrimeRentalsAnalyticsAnnouncementsLogout

Dashboard

Welcome back, Admin!

TOTAL CITIZENS

20

CITY AREAS

5

BUILDINGS

10

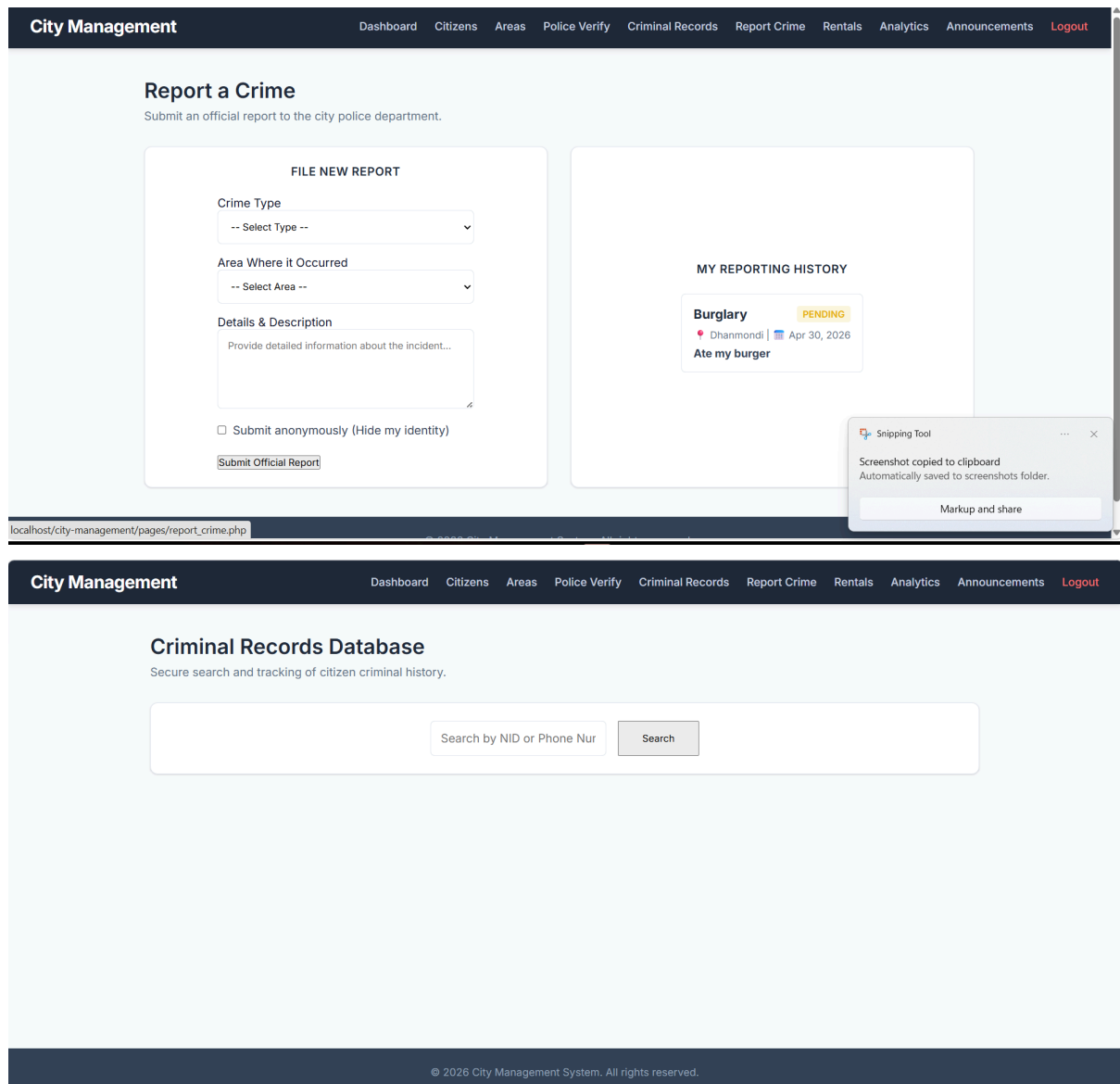
CRIME REPORTS

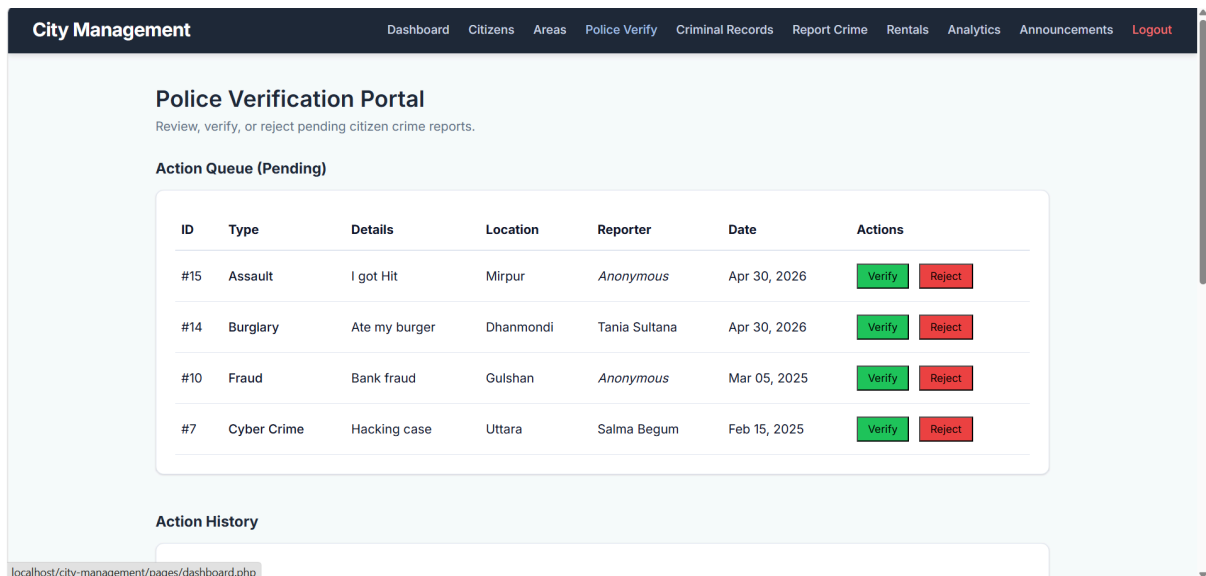
16

© 2026 City Management System. All rights reserved.

Contribution of ID : 24101331, Name : Sadman Rahman

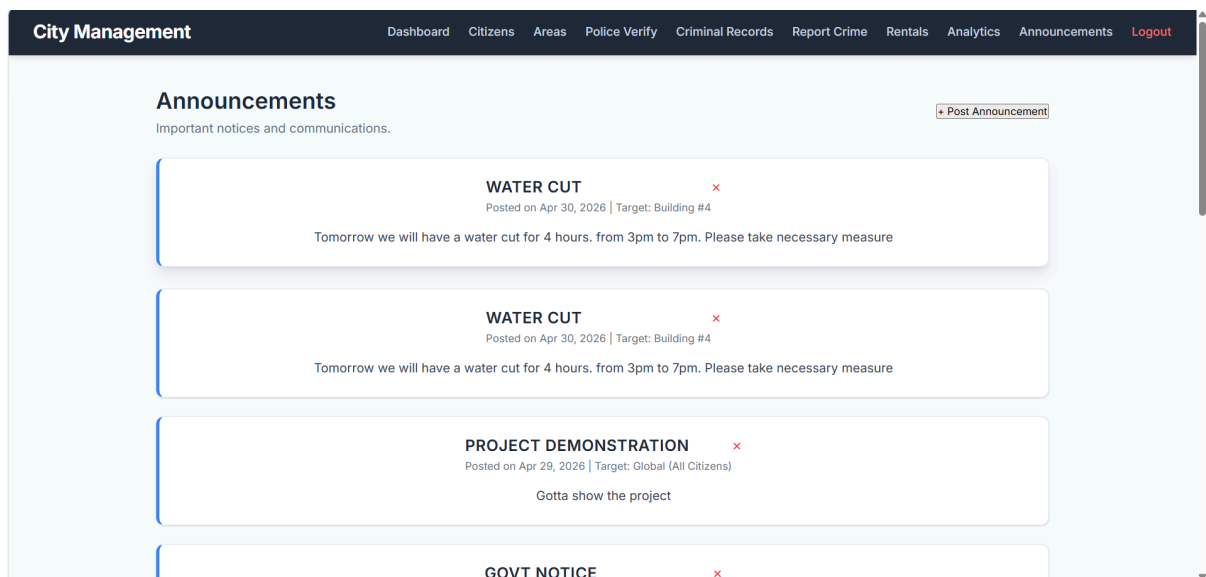
- Criminal History tracking interface
- Crime Reporting system (Anonymous reporting & Verification display)

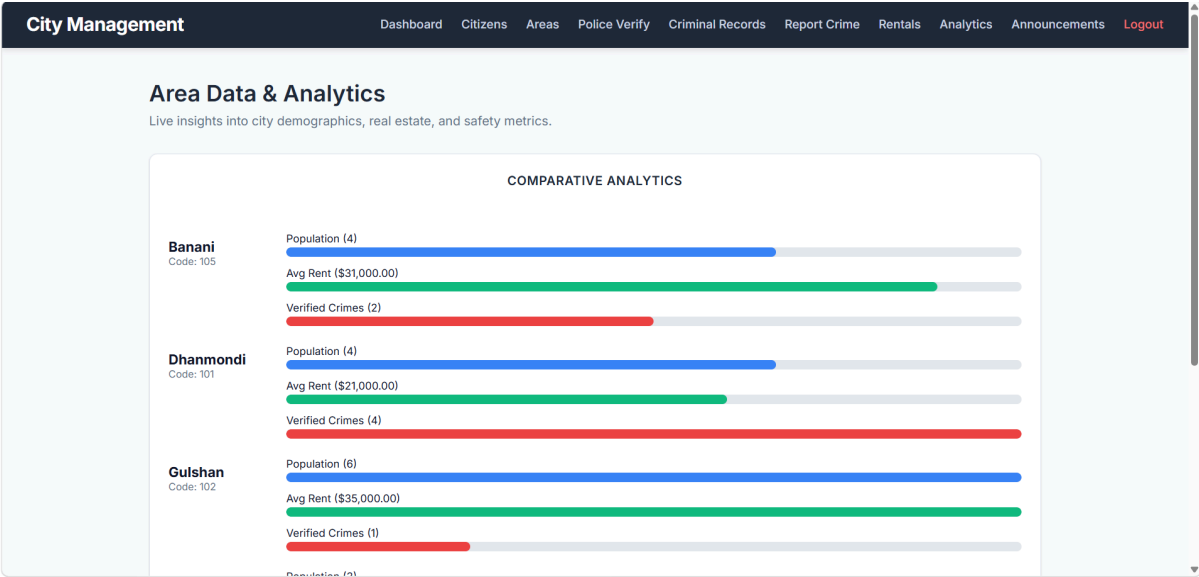




Contribution of ID : 24101328, Name : Oyshi Sarker Nishita

- ☐ Announcement system (cards and posting modal)
- ☐ Area Analytics dashboard with dynamic progress bars for comparison





Backend Development

Briefly discuss about Backend Development and add relevant Screenshots (if required) by mentioning Individual Contributions

The backend was coded with PHP and MySQL, which perform all logical computations, database transactions, and secure data processing. The system relies on session handling for user authentication and role-based access control.

Main features encompass data management, criminal activity tracking, announcement filtration, and analytics calculations. SQL queries with aggregate functions like COUNT() and AVG() were utilized for analyzing data, whereas prepared statements such as prepare() and bind_param() safeguard from SQL injection.

Contribution of ID : 24101392, Name : Fardin Islam

- ☐ Citizen Management
 - Registration, storing, and updating citizen data
 - Role assignment (Owner/Renter)
- ☐ Area Management
 - Managing areas and buildings
 - Handling ownership and renting relationships

Contribution of ID : 24101331, Name : Sadman Rahman

- ☐ Criminal History Tracking
 - Creating and updating citizen crime records
- ☐ Crime Reporting System
 - Anonymous crime reporting
 - Verification system for police approval

Contribution of ID : 24101328, Name : Oyshi Sarker Nishita

- ☐ Announcement Management
 - Role-based posting (Admin and Owner)
 - Targeted announcements (Global, Area, Building)
- ☐ Area Data Analysis & Comparison
 - Conducted area-based data analysis (population, rent, crime)
 - Results updated and visualized in real-time

Source Code Repository

Upload source code to GitHub or Google Drive and share a publicly accessible link in this section of the report.

Urban Citizen Administration System

Conclusion

In conclusion, the City Management System is a step toward making city governance. It brings together in one place the management of citizens, infrastructure, and public safety.

This project aims to end the handling of data in different ways. It wants to give authorities an efficient system. This system lets them access and manage city information at any time. This makes things more transparent. Helps with decision-making.

While building the system, we faced some challenges. One challenge was making sure that user roles are verified accurately. Another challenge was keeping data safe across many levels of access. If not handled well, these limitations can cause risks. Risks like people getting access they should not have, or data being inconsistent.

Plans are geared towards increasing security. Methods such as authentication and encryption will be utilized. Plans are also set to incorporate advanced features. These include timely updates and cloud storage capabilities, which allow the information to scale with us.

The system is expected to evolve to incorporate functionalities that pertain to cities. Examples include Internet of Things integration and automated alarm systems.

In summary, the project serves as an enabling platform for smart cities. The project is evidence of our commitment to continually innovate and adapt to meet the requirements of city administration.

References

- PHP Full Course by Bro Code:
<https://www.youtube.com/watch?v=zZ6vybT1HQs>
- Learn CSS in 1 hour by Bro Code:
<https://www.youtube.com/watch?v=wRNinF7YQqQ>